

# Increased activity of antagonists of growth hormone-releasing hormone substituted at positions 8, 9, and 10

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Antagonists of human growth hormone-releasing hormone (hGHRH) with increased potency and improved enzymatic and chemical stability are needed for potential clinical applications. We synthesized 21 antagonistic analogs of hGHRH(1–29)NH<sub>2</sub>, substituted at positions 8, 9, and 10 of the common core sequence {phenylacetyl-Tyr<sup>1</sup>, D-Arg<sup>2,28</sup>, *para*-chloro-phenylalanine 6, Arg<sup>9</sup>/homoarginine 9, Tyr<sup>10</sup>/O-methyltyrosine 10,  $\alpha$ -aminobutyric acid 15, norleucine 27, Har<sup>29</sup>} hGHRH(1–29)NH<sub>2</sub>. Inhibitory effects on hGHRH-induced GH release were evaluated *in vitro* in a superfused rat pituitary system, as well as *in vivo* after i.v. injection into rats. The binding affinities of the peptides to pituitary GHRH receptors were also determined. Introduction of *para*-amidinophenylalanine 10 yielded antagonists JV-1–62 and -63 with the highest activities *in vitro* and lowest receptor dissociation constants ( $K_i$  = 0.057–0.062 nM). Antagonists JV-1–62 and -63 also exhibited the strongest effect *in vivo*, significantly ( $P$  < 0.05–0.001) inhibiting hGHRH-induced GH release for at least 1 h. *Para*-aminophenylalanine 10 and O-ethyltyrosine 10 substitutions yielded antagonists potent *in vitro*, but His<sup>10</sup>, 3,3'-diphenylalanine 10, 2-naphthylalanine 10, and cyclohexylalanine 10 modifications were detrimental. Antagonists containing citrulline 9 (in MZ-J-7–72), amidinophenylalanine 9 (in JV-1–65), His<sup>9</sup>, D-Arg<sup>9</sup>, citrulline 8, Ala<sup>8</sup>, D-Ala<sup>8</sup>, or  $\alpha$ -aminobutyric acid 8 substituents also had high activity and receptor affinity *in vitro*. However, *in vitro* potencies of analogs with substitution in position 9 correlated poorly with acute endocrine effects *in vivo*, as exemplified by the weak and/or short inhibitory actions of antagonists JV-1–65 and MZ-J-7–72 on GH release *in vivo*. Nevertheless, antagonist JV-1–65 was more potent than JV-1–63 in tests on inhibition of the growth of human prostatic and lung cancer lines xenografted into nude mice. This indicates that oncological activity may be based on several mechanisms. hGHRH antagonists with improved efficacy could be useful for treatment of cancers that depend on insulin-like growth factors or GHRH.

Growth hormone-releasing hormone (GHRH) is a hypothalamic peptide that regulates the secretion of GH from the pituitary by an action exerted on the hypophyseal receptors for GHRH (1, 2). Antagonistic analogs of human GHRH (hGHRH) were designed to block the secretion of GH from the pituitary by inhibiting the binding of hypothalamic GHRH to GHRH receptors. A fall in secretion of GH leads to a decrease in production of hepatic insulin-like growth factor I (IGF-I). Antagonists of GHRH, which inhibit the endocrine pituitary GH/hepatic IGF-I axis, may find use in the treatment of endocrine conditions caused by excessive production of GH, such as acromegaly (1–3). However, the main clinical applications of GHRH antagonists are envisioned to be in the field of cancer (1, 2). Thus, GHRH antagonists could inhibit the growth of malignancies that depend on IGF-1 by suppressing serum levels of IGF-I. Even more important would be the direct interaction of GHRH antagonists with the splice variants (SV) of receptors for GHRH on tumors with the resulting inhibition of tumoral synthesis of IGF-I/II and/or the blocking of the actions of autocrine/paracrine GHRH, which is a growth factor for some tumors (1, 2). Previously, antagonists JV-1-36 and -38 (4), as well

as earlier antagonists MZ-4–71 (5) and MZ-5–156 (6), have been shown to exert antiproliferative effects on a wide range of experimental cancers *in vivo* and *in vitro* (1, 2). However, the activity of these analogs has to be increased to ensure clinical efficacy.

Many agonistic and antagonistic analogs of GHRH, based on the shortest bioactive sequence of hGHRH(1–29)NH<sub>2</sub>, have been synthesized to elucidate the structure–activity relationships and to increase their biological activity (4–13). The first reported GHRH antagonist, termed herein as the “standard antagonist,” has the chemical structure of [Ac-Tyr<sup>1</sup>, D-Arg<sup>2</sup>]hGHRH(1–29)NH<sub>2</sub> (7). This antagonist has been evaluated experimentally and clinically, and in large doses, it reduced GH levels in a patient with acromegaly (3). Most other GHRH antagonists, prepared subsequently, contain a D-Arg<sup>2</sup> substitution, which is necessary for antagonistic activity, in combination with other substituents that increase the receptor-binding affinity and enhance the metabolic stability of the analogs (4–6, 8, 9). Thus, potent and long-acting GHRH antagonists MZ-5–156, JV-1–36, and JV-1–38 contain a phenylacetyl group at the N terminus, together with *para*-chloro-phenylalanine 6,  $\alpha$ -aminobutyric acid 15 (Abu<sup>15</sup>), and norleucine 27 substitutions in the midchain (4–6). The C terminus in analogs JV-1–36 and -38 contains the D-Arg<sup>28</sup>-Har<sup>29</sup>-NH<sub>2</sub> structure (Har, homoarginine) (4). It was also found that replacement of the native Ser<sup>9</sup> by Arg or Har produces antagonists with increased activity and binding affinity to the pituitary receptors for GHRH (4).

Nevertheless, GHRH antagonists that are still more potent and longer-acting *in vivo*, as compared to those available at present, are needed for possible clinical applications. In addition, a careful evaluation must be made of direct effects of GHRH antagonists, mediated by the tumoral SV receptors (14–16) in various tumor models. The antagonist candidates for clinical development should possess high binding affinities and exert biological effects on both the pituitary and the tumoral SV receptors for GHRH. The purpose of the present study was to synthesize a series of antagonists substituted at positions 8, 9, and 10, and to investigate their antagonistic potencies on the rat pituitary receptors, under *in vitro* and *in vivo* conditions. In a parallel effort, the analogs were investigated in tumor models and their antagonistic activities on tumoral SV receptors characterized.

Abbreviations: Alloc, allyloxycarbonyl; Amp, *para*-amidinophenylalanine; Cit, citrulline; GH, growth hormone; Har, homoarginine; hGHRH, human GH-releasing hormone; IGF-I (-II), insulin-like growth factor I (II); SV, splice variant;  $K_i$ , dissociation constant of the inhibitor–receptor complex; Tyr(Et), O-ethyltyrosine; Tyr(Me), O-methyltyrosine.

<sup>†</sup>Tulane University has applied for a patent on the GHRH antagonists listed in this article; J.L.V., A.V.S., and M.Z. are coinventors on that patent.

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**Table 1. Structure of GHRH antagonists with substitutions in the common core sequence [PhAc-Tyr<sup>1</sup>, D-Arg<sup>2</sup>, Phe(pCl)<sup>6</sup>, Abu<sup>15</sup>, Nle<sup>27</sup>, D-Arg<sup>28</sup>, Har<sup>29</sup>]hGHRH(1–29)NH<sub>2</sub>**

| No. | Code no.  | Substitution at position |            |                        |
|-----|-----------|--------------------------|------------|------------------------|
|     |           | 8                        | 9          | 10                     |
| 1   | JV-1-62   | –                        | Arg        | Amp                    |
| 2   | JV-1-63   | –                        | Har        | Amp                    |
| 3   | JV-1-64   | –                        | Arg        | His                    |
| 4   | JV-1-92   | –                        | Har        | 3-Pal                  |
| 5   | JV-1-88   | –                        | Har        | Phe(pNH <sub>2</sub> ) |
| 6   | JV-1-91   | –                        | Har        | Phe(pNO <sub>2</sub> ) |
| 7   | JV-1-93   | –                        | Har        | Tyr(Et)                |
| 8   | JV-1-87   | –                        | Har        | Dip                    |
| 9   | JV-1-86   | –                        | Har        | 2-Nal                  |
| 10  | MZ-J-7-76 | –                        | Arg        | Cha                    |
| 11  | JV-1-66   | –                        | D-Arg      | Tyr(Me)                |
| 12  | JV-1-68   | –                        | Amp        | –                      |
| 13  | JV-1-65   | –                        | Amp        | Tyr(Me)                |
| 14  | JV-1-67   | –                        | His        | Tyr(Me)                |
| 15  | JV-1-69   | –                        | Amp(Alloc) | –                      |
| 16  | MZ-J-7-72 | –                        | Cit        | –                      |
| 17  | MZ-J-7-88 | Ala                      | Arg        | –                      |
| 18  | MZ-J-7-89 | D-Ala                    | Arg        | –                      |
| 19  | MZ-J-7-90 | Abu                      | Arg        | –                      |
| 20  | MZ-J-7-74 | Cit                      | Arg        | –                      |
| 21  | MZ-J-7-78 | Cit                      | Cit        | –                      |
| 22  | JV-1-36*  | –                        | Arg        | –                      |
| 23  | JV-1-38*  | –                        | Har        | Tyr(Me)                |

3-Pal, 3-pyridylalanine; Dip, 3,3'-diphenylalanine; 2-Nal, 2-naphthylalanine; PhAc, phenylacetyl; Phe(pCl), para-chloro-phenylalanine; Phe(pNH<sub>2</sub>), para-amino-phenylalanine; Cha, cyclohexyl alanine; Phe(pNO<sub>2</sub>), para nitro-phenyl alanine. –, the same substituent as in the native hGHRH sequence, namely Asn<sup>8</sup> or Tyr<sup>10</sup>.

\*Reference compounds reported in ref. 4.

## Materials and Methods

**Peptide Synthesis, Purification, and Analysis.** GHRH antagonists were prepared by standard procedures of solid-phase peptide synthesis, as described (4–6). Protected *para*-amidinophenylalanine (Amp) derivative Boc-Amp(Alloc)-OH or Fmoc-Amp(Alloc)-OH (where Boc is *tert*-butoxycarbonyl, Alloc is allyloxycarbonyl, and Fmoc is fluorenylmethoxycarbonyl) was obtained from RSP Amino Acids DBA (Shirley, MA). The other amino acid derivatives, in Boc protected form, as well as resins and reagents used were obtained from Bachem or Advanced Chemtech. All peptides were constructed with amidated C terminus on *p*-methylbenzhydrylamine resin and deprotected and cleaved from the resin with anhydrous hydrogen fluoride (HF), as described (4). The Alloc-protecting group of Amp was removed from the resin-bound peptides (peptides 1, 2, 12, and 13 in Table 1) before HF cleavage, by treatment for 2 h with tetrakis(triphenylphosphine)palladium (0) (3 equivalent) in the presence of chloroform–acetic acid–*N*-methylmorpholine (37:2:1 vol/vol) solvent, under argon atmosphere (17). To prepare peptide 15 (Table 1) with an Amp(Alloc) substituent, the procedure of Alloc removal was omitted and the peptide-resin containing Amp(Alloc) substituent was subjected to hydrogen fluoride treatment. After purification of the crude peptides by semipreparative HPLC (4), the purity of peptides was examined by analytical HPLC (4) and fractions with a purity exceeding 95% were pooled and lyophilized. Molecular masses were confirmed by electrospray mass spectrometry.

**Evaluation of GHRH Antagonistic Activity *in Vitro*.** Antagonistic activities of the peptides on GH release were determined by using

a superfused rat pituitary cell system (18, 19). At the beginning and the end of each superfusion, the membrane-depolarizing K<sup>+</sup> (25 mM KCl) was administered to obtain standard GH releases and evaluate the amount of releasable GH in the cells. After the initial pulse of K<sup>+</sup>, 3-min pulses of 1 nM hGHRH(1–29)NH<sub>2</sub> (“GHRH”) were applied twice. Following these pulses, the antagonistic peptides were perfused through the cells for 9 min at various concentrations (10–30 nM). After this 9-min superfusion, the cells were immediately exposed to a mixture of the GHRH antagonists and 1 nM GHRH for an additional 3 min (0 min response). To check the duration of the antagonistic effect of the analogs, 1 nM GHRH was applied 30, 60, 90, and 120 min later for 3 min (30-, 60-, 90-, and 120-min responses). GH concentration in the 1-ml (3-min) fractions collected was determined by RIA for rat GH. Net integral values of the GH responses were evaluated with a special computer program (18). GH responses at various time points after the administration of the antagonists were compared to the average of the two initial GH responses induced by 1 nM GHRH (reference response), and the inhibitory activities of the antagonists were expressed as percent inhibition of the reference GH response.

**Receptor Binding.** Preparation of rat pituitary membrane fractions and receptor binding of GHRH were performed as previously described (20), by using a sensitive *in vitro* ligand competition assay based on binding of <sup>125</sup>I-labeled [His<sup>1</sup>, Nle<sup>27</sup>]hGHRH(1–32)NH<sub>2</sub> (Nle, norleucine) to rat anterior pituitary membrane homogenates. Briefly, in competitive binding analysis, [His<sup>1</sup>, <sup>125</sup>I-Tyr<sup>10</sup>, Nle<sup>27</sup>]hGHRH(1–32)NH<sub>2</sub> (0.2 nM) was displaced by GHRH antagonists at 10<sup>–6</sup> to 10<sup>–12</sup> M. The final binding affinities were expressed as the dissociation constant of the inhibitor–receptor complex (*K<sub>i</sub>*) and were calculated by using the LIGAND PC computerized curve-fitting program of Munson and Rodbard as modified by McPherson (21). Relative affinities compared to [Ac-Tyr<sup>1</sup>, D-Arg<sup>2</sup>]hGHRH(1–29)NH<sub>2</sub> (standard antagonist) were calculated as the ratio of *K<sub>i</sub>* of the standard antagonist to the *K<sub>i</sub>* of the tested GHRH antagonists.

**GHRH Antagonistic Activity *in Vivo*.** The potency and duration of antagonistic effect of the analogs was tested *in vivo* in young male Sprague–Dawley rats (200–250 g of body weight). The antagonists (80 μg/kg) and hGHRH(1–29)NH<sub>2</sub> (3 μg/kg) were dissolved in 5.5% sterile mannitol and 0.9% NaCl, respectively, given i.v. into the jugular vein of rats under sodium pentobarbital anesthesia. In one experiment, five groups of seven animals each were used. The time elapsed between the administration of the antagonist and subsequent GHRH injection varied among groups (5, 15, 30, and 60 min). Blood samples (0.4 ml) were taken for RIA of GH before the administration of the antagonist (for measurement of the baseline level = GH<sub>baseline</sub>) and 5 min after injection of GHRH (for measurement of the poststimulus level = GH<sub>stimulated</sub>). The controls received mannitol instead of the antagonist, and the GHRH stimulus was given 5 min later. The effect of antagonist was considered significant at a certain time point of *x* min (*x* = 5, 15, 30, or 60) if there was a statistically significant difference between the GH<sub>stimulated(*x* min)</sub> levels after treatment with antagonist and the GH<sub>stimulated(control)</sub> levels in the control group. Statistical evaluation was done by one-way ANOVA followed by Bonferroni's *t* test. The inhibition (percent) at various time points after the administration of the antagonist was calculated by using the formula:

$$\text{Inhibition}_{(x \text{ min})} (\%) =$$

$$\frac{[\text{GH}_{\text{stimulated}(\text{control})} - \text{GH}_{\text{stimulated}(x \text{ min})}]}{[\text{GH}_{\text{stimulated}(\text{control})} - \text{GH}_{\text{baseline}(x \text{ min})}]} \times 100$$

This formula assumes that if the inhibitory activity of an antagonist would be total or 100%, the mean value of GH<sub>stimulated</sub> levels would remain the same as the mean

**Table 2. Inhibitory effects of GHRH antagonists on the GHRH-induced GH release in superfused rat pituitary system**

| Antagonist           |                      |          | % inhibition of GH release |        |        |        |         |
|----------------------|----------------------|----------|----------------------------|--------|--------|--------|---------|
| No.                  | Code no.             | Dose, nM | 0 min                      | 30 min | 60 min | 90 min | 120 min |
| Standard antagonist* |                      | 100      | 52                         | 13     | 0      | 0      | 0       |
| 1                    | JV-1-62              | 30       | 82                         | 89     | 93     | 93     | 92      |
|                      |                      | 10       | 40                         | 76     | 66     | 68     | 47      |
|                      |                      | 30       | 93                         | 99     | 98     | 96     | 97      |
| 2                    | JV-1-63              | 10       | 43                         | 69     | 63     | 64     | 63      |
|                      |                      | 30       | 72                         | 57     | 31     | 24     | ND      |
| 4                    | JV-1-92              | 30       | 58                         | 71     | 74     | 64     | 70      |
| 5                    | JV-1-88              | 30       | 67                         | 88     | 85     | 85     | 81      |
| 6                    | JV-1-91              | 30       | 37                         | 46     | 64     | 63     | 64      |
| 7                    | JV-1-93              | 30       | 53                         | 84     | 86     | 85     | 84      |
| 8                    | JV-1-87              | 30       | -4                         | 10     | -12    | 3      | -20     |
| 9                    | JV-1-86              | 30       | 9                          | 43     | 20     | 34     | 28      |
| 10                   | MZ-J-7-76            | 30       | 3                          | 11     | 22     | 3      | 18      |
| 11                   | JV-1-66              | 30       | 76                         | 87     | 79     | 70     | 64      |
| 12                   | JV-1-68              | 30       | 63                         | 78     | 67     | 71     | 67      |
| 13                   | JV-1-65              | 30       | 72                         | 85     | 82     | 80     | 77      |
| 14                   | JV-1-67              | 30       | 91                         | 88     | 82     | 82     | 75      |
| 15                   | JV-1-69              | 30       | 73                         | 75     | 61     | 58     | 52      |
| 16                   | MZ-J-7-72            | 30       | 84                         | 88     | 100    | 86     | 86      |
| 17                   | MZ-J-7-88            | 30       | 72                         | 90     | 86     | 81     | 75      |
| 18                   | MZ-J-7-89            | 30       | 82                         | 80     | 57     | 34     | 52      |
| 19                   | MZ-J-7-90            | 30       | 41                         | 75     | 79     | 77     | 50      |
| 20                   | MZ-J-7-74            | 30       | 100                        | 87     | 67     | 57     | 63      |
| 21                   | MZ-J-7-78            | 30       | 69                         | 71     | 72     | 54     | 37      |
| 22                   | JV-1-36 <sup>†</sup> | 30       | 64                         | 79     | 75     | 71     | 71      |
|                      |                      | 10       | 20                         | 58     | 40     | 36     | 15      |
| 23                   | JV-1-38 <sup>†</sup> | 30       | 41                         | 68     | 72     | 66     | 69      |

ND, not determined.

\*From ref. 6.

<sup>†</sup>Reference compounds.

GH<sub>baseline</sub> value was in that group. However, if the antagonist had no effect (0% inhibition), the mean value of GH<sub>stimulated</sub> levels in that group would reach the GH<sub>stimulated</sub> value of the control group.

All experiments were performed in accordance with institutional ethical guidelines for the care and use of experimental animals.

**RIA for GH.** Rat GH levels in aliquots of superfusion samples and in serum were measured by double-antibody RIA by using materials supplied by the National Hormone and Pituitary Program, Rockville, MD (rat GH-RP-2/AFP-3190B, rat GH-I-6/AFP-5676B, and anti-rat GH-RIA-5/AFP-411S). Interassay variation was <15% and intraassay variation was <10%.

## Results

**Peptide Design and Synthesis.** In a search for GHRH antagonists with increased potency, 21 analogs of hGHRH(1-29)NH<sub>2</sub> were prepared by solid-phase synthesis (Table 1). All peptides were based on the common core sequence [phenylacetyl-Tyr<sup>1</sup>, D-Arg<sup>2</sup>, *para*-chloro-phenylalanine 6, Arg<sup>9</sup>/Har<sup>9</sup>, Tyr<sup>10</sup>/O-methyltyrosine 10 [Tyr(Me)<sup>10</sup>], aminobutyric acid 15, norleucine 27, D-Arg<sup>28</sup>, Har<sup>29</sup>]hGHRH(1-29)NH<sub>2</sub>, which is present in highly potent GHRH antagonists JV-1-36 and -38 (4). Within this core sequence, JV-1-36 contained Arg<sup>9</sup> and Tyr<sup>10</sup>, whereas JV-1-38 contained Har<sup>9</sup> and Tyr(Me)<sup>10</sup> substitutions. All of the new analogs synthesized, with the exception of **21**, were different in only one position from the reference compounds JV-1-36 or -38 (Table 1). Consequently, the relationship between the structure and activity could be deduced by comparing the biological activity of each peptide to the reference compound to which it was structurally more closely related.

Analog **1-10** contained new substituents in position 10. Thus, peptides **1-5** contained aromatic amino acids of varying basicity,

namely *para*-amidino-phenylalanine 10 (Amp<sup>10</sup>), His<sup>10</sup>, 3-pyridylalanine 10, and *para*-amino-phenylalanine 10. Peptides **6** and **7** contained *para*-nitro-phenylalanine 10 and *O*-ethyltyrosine 10 [Tyr(Et)], respectively. Both of these amino acids have non-charged moieties in the *para* position of the aromatic ring. Peptides **8** and **9** contained 3,3'-diphenylalanine 10 (Dip) and 2-naphthylalanine 10 (2-Nal<sup>10</sup>) respectively, both of which incorporated two aromatic rings and thus were bulkier than tyrosine. In addition, both Dip and 2-Nal lacked any substituent in a position analogous to the *para*-hydroxyl group in tyrosine. Peptide **10** contained cyclohexylalanine 10, an amino acid with a bulky cycloaliphatic side chain.

Analog **11-16** had various substituents in position 9, some of which were basic, whereas others were neutral amino acids. Peptide **11** contained the basic D-Arg<sup>9</sup>, and peptides **12-14** had the basic and aromatic Amp<sup>9</sup> and His<sup>9</sup> substituents. The Amp(Alloc)<sup>9</sup> substituent in peptide **15** lacked basic character, because of the presence of the Alloc-protecting group on the amidino moiety of Amp. Peptide **16** contained citrulline 9 (Cit<sup>9</sup>), an analog of arginine devoid of basic character.

Analog **17-20** had Ala<sup>8</sup>, D-Ala<sup>8</sup>, aminobutyric acid 8, and Cit<sup>8</sup> substituents, respectively. Analog **21** contained Cit<sup>8</sup> and Cit<sup>9</sup> and thus it differs in two substituents from JV-1-36 peptide to which it is the most closely related.

## GHRH Antagonistic Activities *in Vitro* and Receptor-Binding Affinities.

The antagonistic potencies of the compounds *in vitro* were estimated from the results of the superfusion assays and receptor binding measurements, both of these two methods providing results that closely paralleled each other. Inhibitory effects of the antagonists on GHRH-induced GH release in a superfused rat pituitary system are shown in Table 2. The results of GHRH receptor-binding assays on some of the analogs are given in Table 3.



**Table 3.  $K_i$  values and relative affinities of GHRH antagonists to membrane receptors on rat anterior pituitary cells**

| Antagonist          |                      | $K_i^*$ , nM | Relative affinity <sup>†</sup> |
|---------------------|----------------------|--------------|--------------------------------|
| No.                 | Code no.             |              |                                |
| Standard antagonist |                      | 4.88         | 1                              |
| 1                   | JV-1-62              | 0.057        | 86                             |
| 2                   | JV-1-63              | 0.062        | 79                             |
| 5                   | JV-1-88              | 0.070        | 70                             |
| 6                   | JV-1-91              | 0.091        | 54                             |
| 7                   | JV-1-93              | 0.072        | 68                             |
| 8                   | JV-1-87              | 0.33         | 15                             |
| 9                   | JV-1-86              | 0.14         | 35                             |
| 13                  | JV-1-65              | 0.089        | 55                             |
| 14                  | JV-1-67              | 0.090        | 54                             |
| 16                  | MZ-J-7-72            | 0.074        | 66                             |
| 17                  | MZ-J-7-88            | 0.069        | 71                             |
| 18                  | MZ-J-7-89            | 0.081        | 60                             |
| 19                  | MZ-J-7-90            | 0.071        | 69                             |
| 20                  | MZ-J-7-74            | 0.070        | 70                             |
| 21                  | MZ-J-7-78            | 0.073        | 67                             |
| 22                  | JV-1-36 <sup>‡</sup> | 0.079        | 62                             |
| 23                  | JV-1-38 <sup>‡</sup> | 0.085        | 57                             |

\*Dissociation constant of the inhibitor–receptor complex. Values were calculated from duplicate or triplicate determinations.

<sup>†</sup>Expressed relative to [Ac-Tyr<sup>1</sup>, D-Arg<sup>2</sup>] hGHRH(1–29)NH<sub>2</sub> (standard antagonist) = 1.0.

<sup>‡</sup>Reference compounds.

In the series of peptides with substitutions at position 10, analogs **1** and **2**, containing Amp<sup>10</sup>, showed the highest antagonistic potency. These antagonists were tested in the superfusion system at 10 and 30 nM concentrations, and their inhibitory activities on GH release exceeded those of reference peptides JV-1-36 and -38 (Table 2). Analogs **1** and **2** also showed the highest GHRH receptor-binding activities, the affinity values being 86 and 79 times greater, respectively, than that of the standard antagonist [Ac-Tyr<sup>1</sup>, D-Arg<sup>2</sup>]hGHRH(1–29)NH<sub>2</sub> and also exceeding the binding affinities of reference peptides JV-1-36 and -38 (Table 3). Among other analogs with substitutions in position 10, peptides **5** and **7** were slightly more potent, whereas peptides **4** and **6** were somewhat less potent than antagonists JV-1-36 and -38. Peptides **3**, **8**, **9**, and **10** had reduced or greatly reduced inhibitory activity on GH release and GHRH receptor affinity, as compared to antagonists JV-1-36 and -38. Our results with analogs substituted at position 10 indicate that certain *para*-substituted aromatic amino acids are preferred embodiments, as seen from the high activity of antagonists containing Amp<sup>10</sup>, *para*-amino-phenylalanine 10 [Phe(pNH<sub>2</sub>)<sup>10</sup>], and Tyr(Et)<sup>10</sup> substitutions. Although Amp is a basic amino acid, positively charged amino acids in general do not seem to be favorable in position 10, because the Phe(pNH<sub>2</sub>)<sup>10</sup> and Tyr(Et)<sup>10</sup> substitutions, which are not charged at physiological pH, led to a higher potency than a His<sup>10</sup> replacement. The very substantial decrease in antagonistic activity produced by 3,3'-diphenylalanine 10 and 2-naphthylalanine 10 replacements could be linked to the bulkiness of their side chains and perhaps their lack of a *para* substituent. In addition, the low activity of the cyclohexylalanine 10 analog indicates that the substituent at position 10 should have an aromatic character.

Among the analogs substituted in position 9, peptide **16** had a clearly stronger antagonistic effect (Table 2) and slightly higher receptor-binding affinity (Table 3) than reference peptides JV-1-36 and -38 or other analogs substituted at this position. Based on superfusion and receptor-binding results, peptides **11**, **12**, **13**, and **14** were about as potent, whereas peptide **15** was slightly less potent than the reference antagonists JV-1-36 and -38. Thus, in the series substituted at position 9, the introduction of Cit<sup>9</sup>, lacking a positive charge on its side chain, produced the

most potent antagonist. In addition, D-Arg, Amp, His, and Amp(Alloc) substituents were all favorable and the antagonistic effects of peptides containing these amino acids in position 9 were roughly equivalent to the reference peptides JV-1-36 and -38 containing Arg<sup>9</sup> and Har<sup>9</sup>, respectively.

Among the analogs substituted at position 8, peptides **17**, **19**, and **20** displayed a higher receptor-binding affinity (Table 3), and peptides **17** and **20** also exerted a slightly greater inhibition of GH release than reference peptides JV-1-36 and -38 (Table 2). Peptide **18** was about as potent as the reference peptides. Thus, replacement of Asn<sup>8</sup> by Ala<sup>8</sup> or Cit<sup>8</sup> in GHRH antagonists had beneficial effects. Peptide **21**, substituted with Cit at positions 8 and 9, has a potency similar to those of antagonists JV-1-36 and -38, based on the results of superfusion and receptor binding assays (Tables 2 and 3).

**GHRH Antagonistic Activities *in Vivo*.** Some of the antagonists with high activity *in vitro* were also evaluated *in vivo* to assess their potency and duration of action. The results of *in vivo* tests are presented in Table 4, and some of the effects produced are illustrated in Fig. 1 by using the data from two representative experiments. Fig. 1A shows an antagonist that strongly inhibited GH release at 5 min, because the GH<sub>stimulated(5 min)</sub> values were only slightly higher than the GH<sub>baseline</sub> values of that group. The activity of antagonist gradually weakens at later time points, and the GH<sub>stimulated</sub> value in the 60-min group was similar to that observed in the control group, indicating that the antagonist had no inhibitory effect at 60 min (Fig. 1A). Fig. 1B shows an antagonist with a strong and protracted activity, significantly inhibiting GH release *in vivo* even at 60 min after administration. The methods and their underlying principles described in *Materials and Methods*, used for statistical analysis of the effects and for calculating percent inhibition, are different from those used in a previous publication (4), where reference antagonists JV-1-36 and -38 were first reported. We found that these new methods more accurately reflect the findings of the biological system. Results of previous measurement on reference peptides JV-1-36 and -38 are shown in Table 4 as Experiment 1, along with reanalysis of data according to the new protocol. The experiments with JV-1-36 and -38 were also repeated and are shown as Experiment 2 in Table 4.

Among the antagonists tested *in vivo*, **1** (JV-1-62) and **2** (JV-1-63) had the strongest and most protracted inhibitory effect that was significant ( $P < 0.01$ ), even after 60 min (Table 4). Peptide **7** had a weak effect, similar to that of reference antagonist JV-1-38, from which it is structurally derived. The inhibitory effect of analog **7** was significant at 5 min after administration. Peptide **13** (JV-1-65) was tested twice and on both occasions had only a very weak inhibitory effect *in vivo* that was not significant at any time point (Table 4). Peptide **16** had a strong but short-lasting antagonistic effect, and it proved inferior to JV-1-36, from which it is structurally derived. Peptide **20** produced a strong and medium lasting inhibitory effect, similar to that of JV-1-36 (Table 4).

## Discussion

In the search for GHRH antagonists with improved activity, we synthesized and tested biologically a series of GHRH antagonistic analogs substituted at positions 8, 9, and 10. The substitution at position 10 was chosen because it was shown that in the native GHRH Tyr<sup>10</sup> likely participates in receptor contact and is essential for biological activity and the high binding affinity of GHRH to its receptors (10, 12). We speculated that certain Tyr<sup>10</sup> replacements would enhance the binding affinity and possibly increase the antagonistic activity of our analogs. Regarding position 9, it was revealed that the Ser<sup>9</sup> residue of native GHRH is not essential for either the high affinity or the optimal receptor-binding conformation of GHRH. However, replace-

**Table 4. *In vivo* inhibitory effects of GHRH antagonists on the GH release in rats induced by exogenous GHRH**

| Antagonists |                                        |                          | Serum GH, ng/ml, and relative inhibition, percent |                        |                        |                        |                          |
|-------------|----------------------------------------|--------------------------|---------------------------------------------------|------------------------|------------------------|------------------------|--------------------------|
| No.         | Code no.                               | Test                     | Control                                           | 5 min                  | 15 min                 | 30 min                 | 60 min                   |
| 1           | JV-1-62                                | GH <sub>baseline</sub>   | 81.8 ± 12.4                                       | 97.2 ± 19.6            | 76.2 ± 6.96            | 97.6 ± 19.8            | 63.1 ± 14.1              |
|             |                                        | GH <sub>stimulated</sub> | 1,604 ± 339                                       | 310 ± 71.8*            | 412 ± 34.0*            | 778 ± 141 <sup>†</sup> | 648 ± 181 <sup>‡</sup>   |
|             |                                        | Inhibition               |                                                   | 86%                    | 78%                    | 55%                    | 62%                      |
| 2           | JV-1-63                                | GH <sub>baseline</sub>   | 91.7 ± 15.8                                       | 58.1 ± 6.04            | 79.6 ± 8.03            | 74.3 ± 10.3            | 88.3 ± 8.77              |
|             |                                        | GH <sub>stimulated</sub> | 1,971 ± 329                                       | 290 ± 68.5*            | 501 ± 89.8*            | 563 ± 184*             | 1,077 ± 109 <sup>‡</sup> |
|             |                                        | Inhibition               |                                                   | 88%                    | 78%                    | 74%                    | 47%                      |
| 7           | JV-1-93                                | GH <sub>baseline</sub>   | 68.2 ± 16.6                                       | 68.3 ± 15.4            | 66.3 ± 13.5            | 67.4 ± 16.3            | 45.3 ± 13.5              |
|             |                                        | GH <sub>stimulated</sub> | 1,374 ± 134                                       | 793 ± 101 <sup>†</sup> | 938 ± 195              | 969 ± 63.1             | 931 ± 148                |
|             |                                        | Inhibition               |                                                   | 44%                    | 33%                    | 31%                    | 33%                      |
| 13          | JV-1-65<br>(Experiment 1)              | GH <sub>baseline</sub>   | 56.5 ± 11.0                                       | 38.6 ± 5.86            | 112 ± 17.0             | 78.5 ± 32.1            | ND                       |
|             |                                        | GH <sub>stimulated</sub> | 906 ± 86.5                                        | 601 ± 64.2             | 919 ± 112              | 920 ± 123              | ND                       |
|             |                                        | Inhibition               |                                                   | 35%                    | 0%                     | 0%                     |                          |
| 13          | JV-1-65<br>(Experiment 2)              | GH <sub>baseline</sub>   | 109 ± 20.2                                        | 79.0 ± 19.7            | 82.7 ± 24.4            | ND                     | ND                       |
|             |                                        | GH <sub>stimulated</sub> | 1,819 ± 191                                       | 1,551 ± 291            | 1,590 ± 264            | ND                     | ND                       |
|             |                                        | Inhibition               |                                                   | 15%                    | 13%                    |                        |                          |
| 16          | MZ-J-7-72                              | GH <sub>baseline</sub>   | 46.7 ± 7.69                                       | 69.2 ± 24.1            | 67.7 ± 14.6            | 51.6 ± 5.50            | 64.8 ± 15.3              |
|             |                                        | GH <sub>stimulated</sub> | 1,018 ± 100                                       | 241 ± 11.0*            | 829 ± 124              | 996 ± 78.2             | 1,022 ± 156              |
|             |                                        | Inhibition               |                                                   | 82%                    | 20%                    | 2%                     | 0%                       |
| 20          | MZ-J-7-74                              | GH <sub>baseline</sub>   | 62.4 ± 17.7                                       | 41.7 ± 11.2            | 73.0 ± 23.9            | 23.0 ± 7.78            | 96.5 ± 13.2              |
|             |                                        | GH <sub>stimulated</sub> | 1,130 ± 110                                       | 143 ± 40.3*            | 464 ± 107 <sup>‡</sup> | 900 ± 121              | 1,340 ± 194              |
|             |                                        | Inhibition               |                                                   | 91%                    | 63%                    | 21%                    | 0%                       |
| 23          | JV-1-36 <sup>§</sup><br>(Experiment 1) | GH <sub>baseline</sub>   | 158 ± 24                                          | 179 ± 14               | 139 ± 6                | 151 ± 9                | 120 ± 7                  |
|             |                                        | GH <sub>stimulated</sub> | 565 ± 116                                         | 198 ± 22*              | 261 ± 24 <sup>‡</sup>  | 292 ± 19 <sup>‡</sup>  | 479 ± 88                 |
|             |                                        | Inhibition               |                                                   | 95%                    | 71%                    | 66%                    | 19%                      |
| 22          | JV-1-36 <sup>§</sup><br>(Experiment 2) | GH <sub>baseline</sub>   | 48.0 ± 11.0                                       | 39.2 ± 12.2            | 73.1 ± 9.39            | 66.1 ± 9.67            | 73.2 ± 14.5              |
|             |                                        | GH <sub>stimulated</sub> | 1,307 ± 94.9                                      | 180 ± 28.1*            | 689 ± 69.3*            | 1,126 ± 101            | 1,337 ± 154              |
|             |                                        | Inhibition               |                                                   | 89%                    | 50%                    | 15%                    | 0%                       |
| 23          | JV-1-38 <sup>§</sup><br>(Experiment 1) | GH <sub>baseline</sub>   | 51 ± 5                                            | 86 ± 15                | 66 ± 3                 | 83 ± 13                | 76 ± 12                  |
|             |                                        | GH <sub>stimulated</sub> | 377 ± 19                                          | 245 ± 14               | 237 ± 52 <sup>†</sup>  | 371 ± 10               | 387 ± 55                 |
|             |                                        | Inhibition               |                                                   | 45%                    | 45%                    | 2%                     | 0%                       |
| 23          | JV-1-38 <sup>§</sup><br>(Experiment 2) | GH <sub>baseline</sub>   | 42.2 ± 8.53                                       | 65.3 ± 9.71            | 140 ± 18.2             | 149 ± 30.3             | 182 ± 42.9               |
|             |                                        | GH <sub>stimulated</sub> | 1,317 ± 155                                       | 886 ± 139              | 1081 ± 200             | 1,268 ± 135            | 1,555 ± 261              |
|             |                                        | Inhibition               |                                                   | 34%                    | 20%                    | 4                      | 0%                       |

GH levels are mean ± SEM of five to seven animals per group. ND, not determined.

\**P* < 0.001 vs. control.

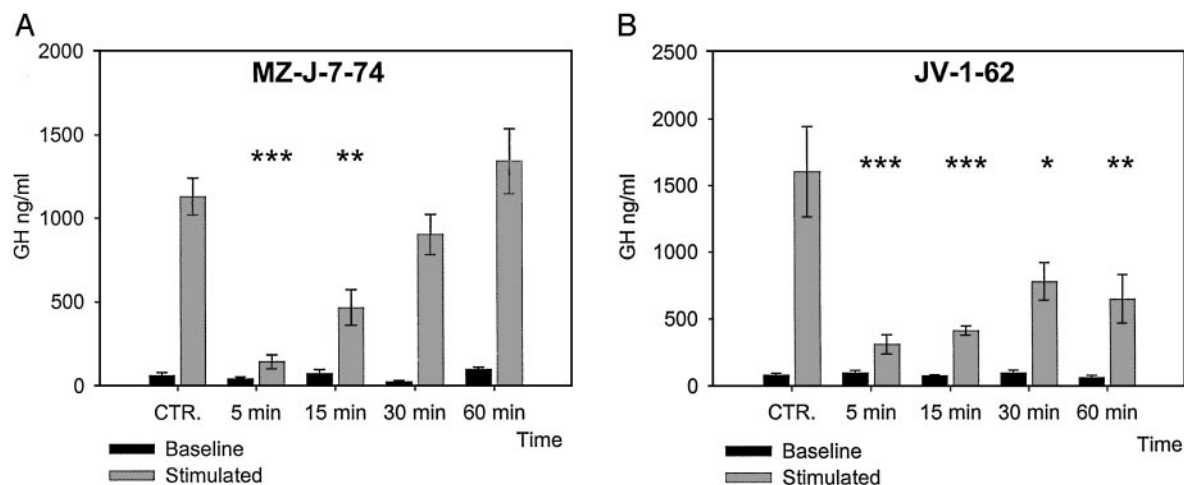
<sup>†</sup>*P* < 0.05 vs. control.

<sup>‡</sup>*P* < 0.01 vs. control.

<sup>§</sup>Reference compounds.

ment of Ser<sup>9</sup> in GHRH antagonists by the basic amino acids Arg<sup>9</sup> or Har<sup>9</sup> enhanced their activity (4). Thus we were interested in whether other amino acid substituents at position 9 could further

enhance the potency of GHRH antagonists and whether a positively charged amino acid is necessary for that effect. Analogs containing basic amino acids are of interest, because it



**Fig. 1.** Inhibition of serum GH levels stimulated by exogenous hGHRH(1–29)NH<sub>2</sub> (3 μg/kg i.v.) in rats *in vivo* by GHRH antagonists MZ-J-7-74 (A) and JV-1-62 (B) (both at 80 μg/kg i.v.), given at various time intervals before GHRH stimulus. Control animals received vehicle solvent instead of antagonist. Baseline levels of GH were measured before the injection of antagonist or vehicle solvent. Vertical bars, SEM. \*, *P* < 0.05 vs. control; \*\*, *P* < 0.01 vs. control; \*\*\*, *P* < 0.001 vs. control.

was observed that positively charged amino acid residues in hormones are sometimes necessary for their high receptor-binding affinities (22, 23). The reason for preparing analogs substituted at position 8 was that GHRH is prone to hydrolytic degradation at the Asn<sup>8</sup> residue (24). Therefore, it would be beneficial to replace Asn<sup>8</sup> with a chemically stable modification in GHRH analogs intended for clinical development.

Several analogs substituted at positions 8–10 were as active *in vitro* or more active than the parent antagonists JV-1-36 and -38. Among the antagonists prepared, peptides **1** (JV-1-62) and **2** (JV-1-63), with Amp<sup>10</sup> substitution, proved to be the most potent. In agreement with their strong antagonistic effect and receptor-binding affinity *in vitro*, their inhibitory effect on GH release *in vivo* proved to be highly significant ( $P < 0.01$ ) 47–62%, 1 h after administration. However, other antagonists that had strong effects *in vitro* turned out to be much less potent *in vivo*. Thus, the effect of peptide **7** (JV-1-93) *in vivo* was only significant at 5 min after administration; the effect of peptide **13** (JV-1-65) was very faint and did not reach significance level at any time point *in vivo*. The weak antagonistic effect *in vivo* of peptide **16** (MZ-J-7-72), as compared to JV-1-36, was also surprising. Peptide **16** contains Cit<sup>9</sup>, whereas JV-1-36 contains Arg<sup>9</sup>; it was expected that the former analog would have a stronger and more protracted effect *in vivo*, because Cit-containing peptides are more resistant to trypsin-like enzymatic degradation in blood than peptides with Arg. The present results are similar to those obtained previously with an early antagonist MZ-4-243, which showed extremely strong antagonistic effects *in vitro*, but it had only weak effect *in vivo* (5). Thus, it appears it is generally not possible to predict which antagonists with high *in vitro* potencies will also exert strong effects *in vivo*. It should be mentioned that a discrepancy also exists between the *in vivo* activities of reference peptides JV-1-36 and -38, JV-1-36 being much more potent in acute tests *in vivo* than JV-1-38 (see Table 4). This discrepancy was partly overlooked when antagonists JV-1-36 and -38 were first reported (4), because a different and less accurate method was used for calculating *in vivo* activities.

More recent results indicate that antagonists with low activity in the acute tests of GH release *in vivo* may surprisingly have strong antiproliferative effects and could even inhibit serum IGF-I levels in chronic experiments. We found recently that antagonist JV-1-65 significantly inhibited the growth of DMS-153 human small-cell lung carcinoma in nude mice, and its activity was higher than that of antagonist JV-1-63 (25). In

addition, serum levels of IGF-I were significantly suppressed to a similar extent by both antagonists (25). In other experiments, the growth of PC-3 human prostate cancers in nude mice was also more effectively suppressed by antagonist JV-1-65 than by JV-1-63 (M. Letsch, A.V.S., J.L.V., and G.H., unpublished work). The results are intriguing because JV-1-65 did not significantly inhibit GH release *in vivo*, in contrast to antagonist JV-1-63, which had a strong effect (Table 4). A similar observation can be made of antagonists JV-1-36 and -38. JV-1-38 has only a low activity in acute *in vivo* tests for inhibition of GH release but is a potent antitumor agent in several cancer models, with oncological effects comparable to those of JV-1-36 (2, 26). The differences in oncological efficacy of GHRH antagonists can be readily explained by their multiple mechanisms of action. Accordingly, in addition to the suppression of the pituitary GH-hepatic IGF-I axis, GHRH antagonists inhibit the proliferation of various cancers by a direct action on tumors. These direct inhibitory effects appear to be mediated by the tumoral SVs of GHRH receptors and in some cancers, GHRH antagonists may block the stimulatory action of tumoral autocrine GHRH. Still, in many other human cancers, GHRH antagonists inhibit the production of tumoral IGF-I and -II by an action presumably exerted through SVs of GHRH receptors. The antitumor effects of the present series of GHRH antagonists must be fully assessed in future experiments. Endocrine studies based on chronic *in vivo* treatment with GHRH antagonists will also help to clarify whether the long-term inhibitory effects of the s.c. injected antagonists on serum IGF-I levels are in accord with their short-term effects on GH release after i.v. injection, as observed in this study.

GHRH antagonists analogs of the type reported here could be beneficial for the treatment of cancers that depend on IGF-I and -II or autocrine tumoral GHRH. In addition, GHRH antagonists could be tried in patients with endocrine disorders characterized by excessive GH production.

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